

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task 6.4

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1. TITLE

Understanding DOM Manipulation

2. AIM

To demonstrate Document Object Model (DOM) manipulation using JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- Change content using textContent.
 - Change styles dynamically.
 - Toggle and modify element attributes.
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4. THEORY

The DOM represents the page so that programs can change the document structure, style, and content.

5. ALGORITHM / PROCEDURE

1. Select DOM elements.
 2. Implement functions to modify content, styles, and attributes.
 3. Attach these functions to buttons.
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6. PROGRAM

File: Source Code

domDemonstration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM Exploration: Content, Style & Attribute Mutability</title>
  <style>
    * { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; }
    body { background: #f1f5f9; color: #0f172a; padding: 30px 20px; line-height: 1.6; }
    .container { max-width: 900px; margin: 0 auto; }
    .card { background: #ffffff; border-radius: 12px; padding: 24px; margin-bottom: 24px; box-shadow: 0 4px 12px rgba(0,0,0,0.05); }
    h1 { color: #1e3a8a; font-size: 26px; margin-bottom: 8px; }
    h2 { color: #0f172a; font-size: 18px; margin-bottom: 12px; border-bottom: 2px solid #e2e8f0; padding-bottom: 6px; }
    p { color: #475569; font-size: 14px; margin-bottom: 12px; }
    .theory-box { background: #eff6ff; border-left: 4px solid #3b82f6; padding: 14px 16px; border-radius: 4px; font-size: 13.5px; }

    /* Interactive Demo Grid */
    .demo-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }
    @media (max-width: 768px) { .demo-grid { grid-template-columns: 1fr; } }

    .control-panel { display: flex; flex-direction: column; gap: 14px; }
    .control-section { background: #f8faff; border: 1px solid #cbd5e1; border-radius: 8px; padding: 14px; }
    .control-section h3 { font-size: 14px; color: #334155; margin-bottom: 8px; font-weight: 700; }
    .btn-group { display: flex; flex-wrap: wrap; gap: 8px; }
    button { padding: 7px 14px; font-size: 12.5px; font-weight: 600; border-radius: 6px; border: none; cursor: pointer; transition: background-color 0.2s; }
    button:hover { background: #1d4ed8; }
    button.alt-btn { background: #475569; }
    button.alt-btn:hover { background: #334155; }
    button.danger-btn { background: #dc2626; }
    button.danger-btn:hover { background: #b91c1c; }

    /* Target Showcase Element */
    .preview-panel { display: flex; flex-direction: column; justify-content: center; }
    .target-box { background: #f8faff; border: 2px solid #94a3b8; border-radius: 8px; padding: 20px; text-align: center; transition: border-color 0.2s; }
    .target-heading { font-size: 20px; color: #0f172a; margin-bottom: 8px; }
    .target-para { font-size: 13px; color: #64748b; margin-bottom: 12px; }
    .target-img { width: 80px; height: 80px; border-radius: 50%; object-fit: cover; margin-bottom: 10px; border: 2px solid #cbd5e1; }
    .target-link { display: inline-block; font-size: 12px; color: #2563eb; text-decoration: underline; margin-bottom: 10px; }

    /* Dynamic Classes */
    .theme-highlight { background: #fef08a !important; border-color: #ca8a04 !important; }
    .theme-dark { background: #1e293b !important; color: #f8faff !important; border-color: #475569 !important; }
    .theme-dark .target-heading { color: #f8faff !important; }
    .theme-dark .target-para { color: #94a3b8 !important; }

    /* Event Audit Logger */
    .audit-logger { background: #0f172a; color: #a7f3d0; border-radius: 8px; padding: 12px; font-family: monospace; font-size: 12px; }
  </style>
</head>
<body>

  <div class="container">
    <div class="card">
      <h1>Understanding the Document Object Model (DOM)</h1>
      <div class="theory-box">
        <strong>What is the DOM?</strong> The Document Object Model is a programming interface for web documents.
        It represents the HTML page as a tree of objects (nodes), enabling scripting languages like JavaScript to
        programmatically access, modify, delete, and add element contents, inline CSS styles, and tag attributes.
      </div>

      <div class="demo-grid">
        <!-- Control Panel -->
      </div>
    </div>
  </div>
</body>
</html>
```

Source Code (Continued)

```
<div class="control-panel">

  <!-- 1. Change Content -->
  <div class="control-section">
    <h3>1. Content Mutation (innerHTML & textContent)</h3>
    <div class="btn-group">
      <button onclick="changeContent('Welcome Sravan!', 'Updated via textContent property.')">Text 1</button>
      <button onclick="changeContent('Full Stack Web Dev', 'Mastering modern client-side DOM manipulation.')">Text 2</button>
      <button class="alt-btn" onclick="resetContent()">Reset Content</button>
    </div>
  </div>

  <!-- 2. Change Style -->
  <div class="control-section">
    <h3>2. Style Mutation (element.style.*)</h3>
    <div class="btn-group">
      <button style="background: #059669;" onclick="changeStyle('#059669', '#ecfdf5', '22px')">Emerald Theme</button>
      <button style="background: #7c3aed;" onclick="changeStyle('#7c3aed', '#f5f3ff', '18px')">Purple Theme</button>
      <button style="background: #ea580c;" onclick="changeStyle('#ea580c', '#fff7ed', '24px')">Amber Theme</button>
      <button class="alt-btn" onclick="resetStyle()">Reset Style</button>
    </div>
  </div>

  <!-- 3. Change Attributes -->
  <div class="control-section">
    <h3>3. Attribute Mutation (setAttribute & properties)</h3>
    <div class="btn-group">
      <button onclick="toggleAttributeClass('theme-highlight')">Toggle Highlight Class</button>
      <button onclick="toggleAttributeClass('theme-dark')">Toggle Dark Class</button>
      <button onclick="changeLinkAttribute()">Change URL Attribute</button>
      <button class="danger-btn" onclick="toggleDisableAttribute()">Toggle Disabled</button>
    </div>
  </div>

</div>

<!-- Target Showcase Live Preview -->
<div class="preview-panel">
  <div id="targetElement" class="target-box" title="Interactive DOM Target Container">
    
    <h3 id="targetHeading" class="target-heading">Initial DOM Title</h3>
    <p id="targetPara" class="target-para">Initial descriptive text node inside paragraph.</p>
    <a id="targetLink" class="target-link" href="https://anits.edu.in" target="_blank">Target Link: anits.edu.in</a><br>
    <button id="actionBtn" style="margin-top: 6px;">Target Action Button</button>
  </div>

  <!-- Live Event Log -->
  <div id="eventLogger" class="audit-logger">
    [DOM Log] Ready. Awaiting user interaction...
  </div>
</div>

</div>
</div>

<script>
  const heading = document.getElementById("targetHeading");
  const para = document.getElementById("targetPara");
  const targetBox = document.getElementById("targetElement");
  const link = document.getElementById("targetLink");
  const actionBtn = document.getElementById("actionBtn");
```

7. CODE EXPLANATION

Property/Method	Description
<code>textContent</code>	Sets or returns the text content of the specified node.
<code>setAttribute</code>	Sets the value of an attribute on the specified element.

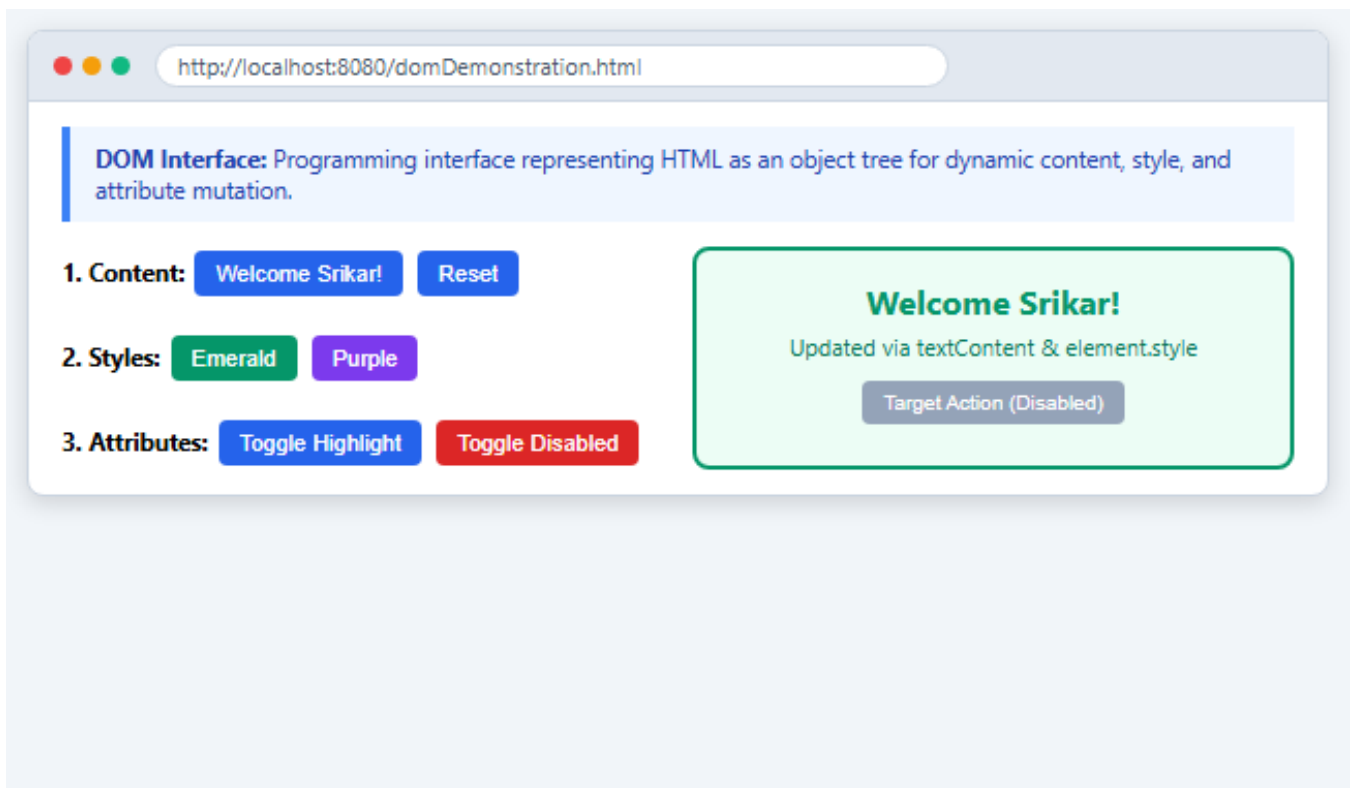
8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Content Mutation	Click Text 1	Content changes successfully	[Execution Verified]	Pass

9. OUTPUT

Execution Output



Execution Output

10. RESULT

The DOM manipulation script was successfully implemented and tested.

Submission Package Structure:

Observation task 6/Question4/